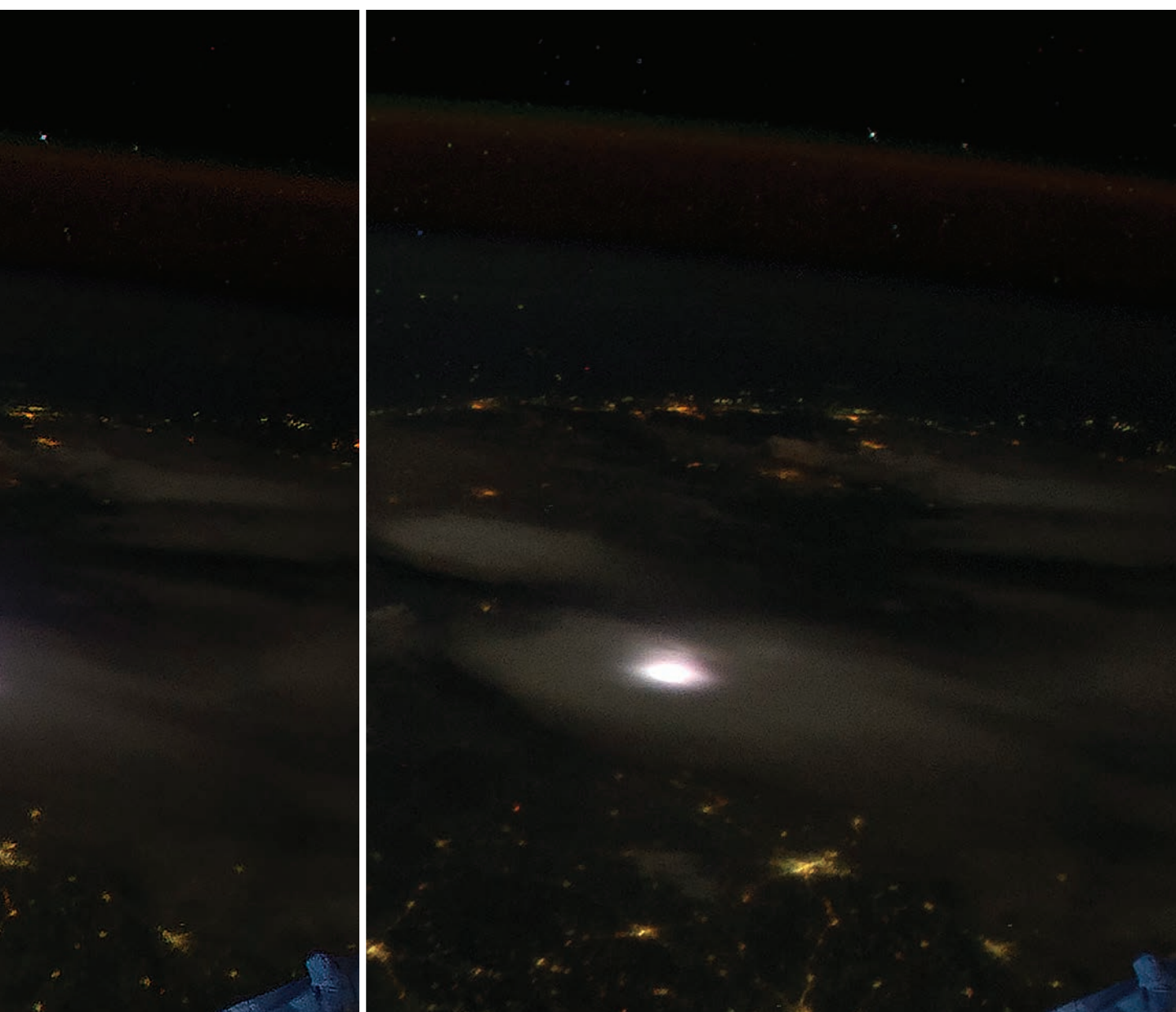
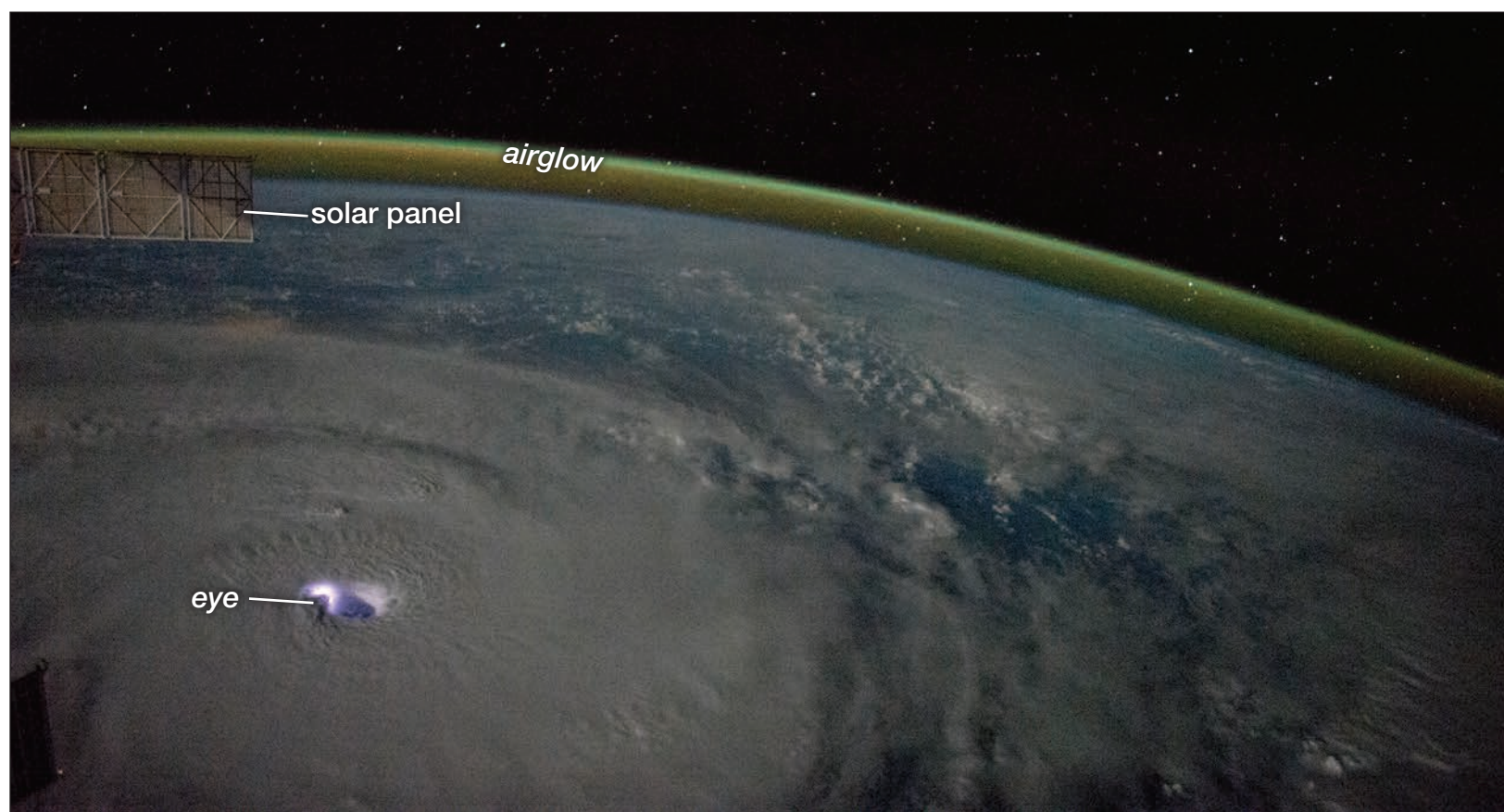
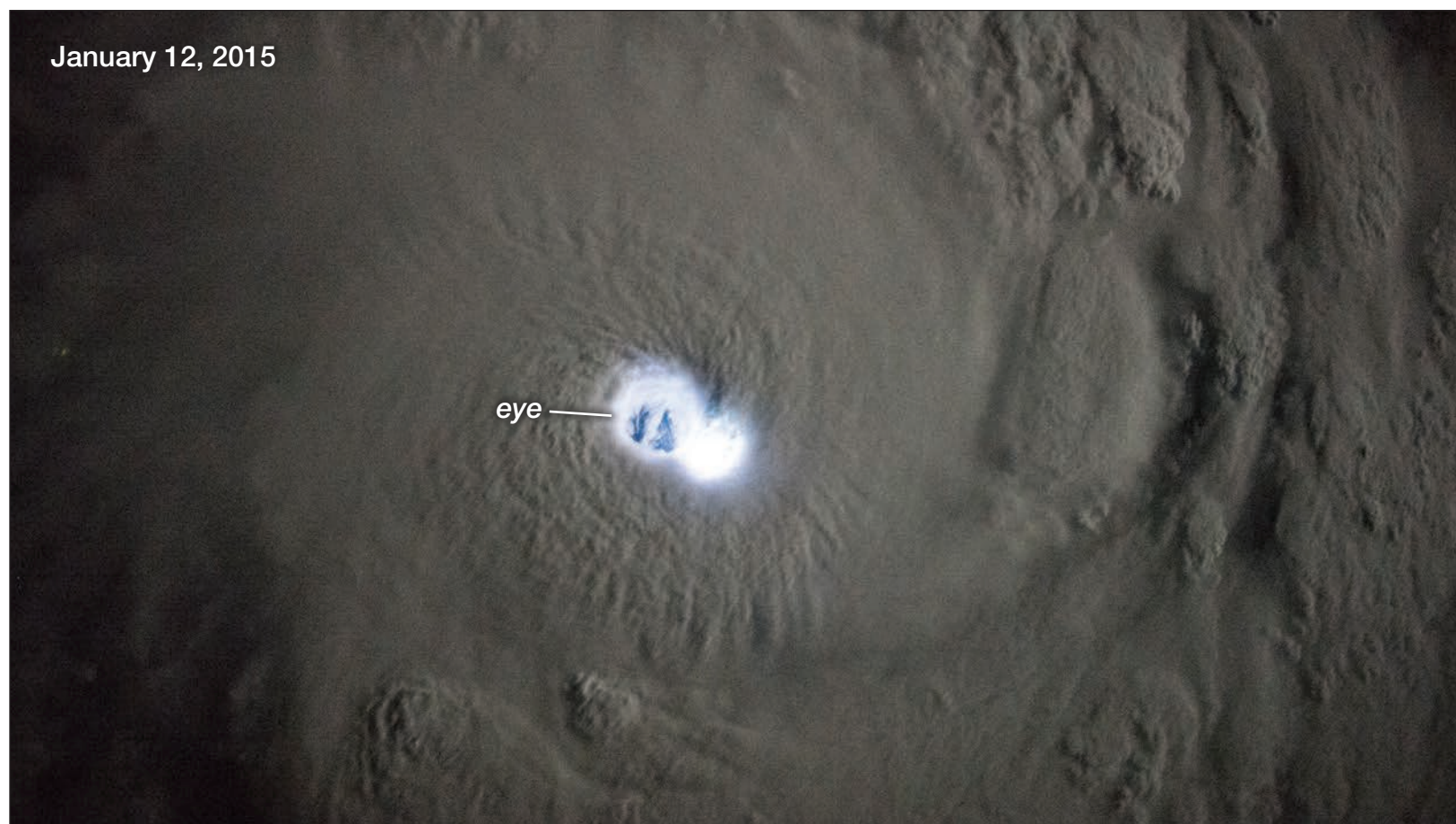


Expedition 31 astronauts on the ISS as they traveled southeast from central Myanmar (Burma) to just north of Malaysia. Red sprites are difficult to observe because they last for just a few milliseconds and occur above thunderstorms—meaning they are usually blocked from view on the ground by the very clouds that produce them. The sprites send pulses of electrical energy up toward the edge of space—into the electrically charged layer known as the ionosphere—instead of down to Earth’s surface. They are rich with radio noise, and can sometimes occur in bunches.







The Electric Eye of Cyclone Bansi—Indian Ocean

Though these images may look like they come from a science fiction movie, they are in fact photographs of tropical cyclone Bansi as seen at night by astronauts on the ISS on January 12, 2015. The images were taken when the ISS was east of Madagascar. The dim swirl of the cloud bands covers the ocean surface in both night images. The eye of the cyclone is brilliantly lit by lightning in or near the eye wall. The low-light settings of the camera used to take the image accentuate the contrast.

In the bottom image, the camera also accentuates the yellow-green airglow above Earth's limb, an atmospheric phenomenon frequently reported by astronauts. Stars appear above the airglow layer, and the solar panels of a docked Russian spacecraft jut into the upper left of the image.

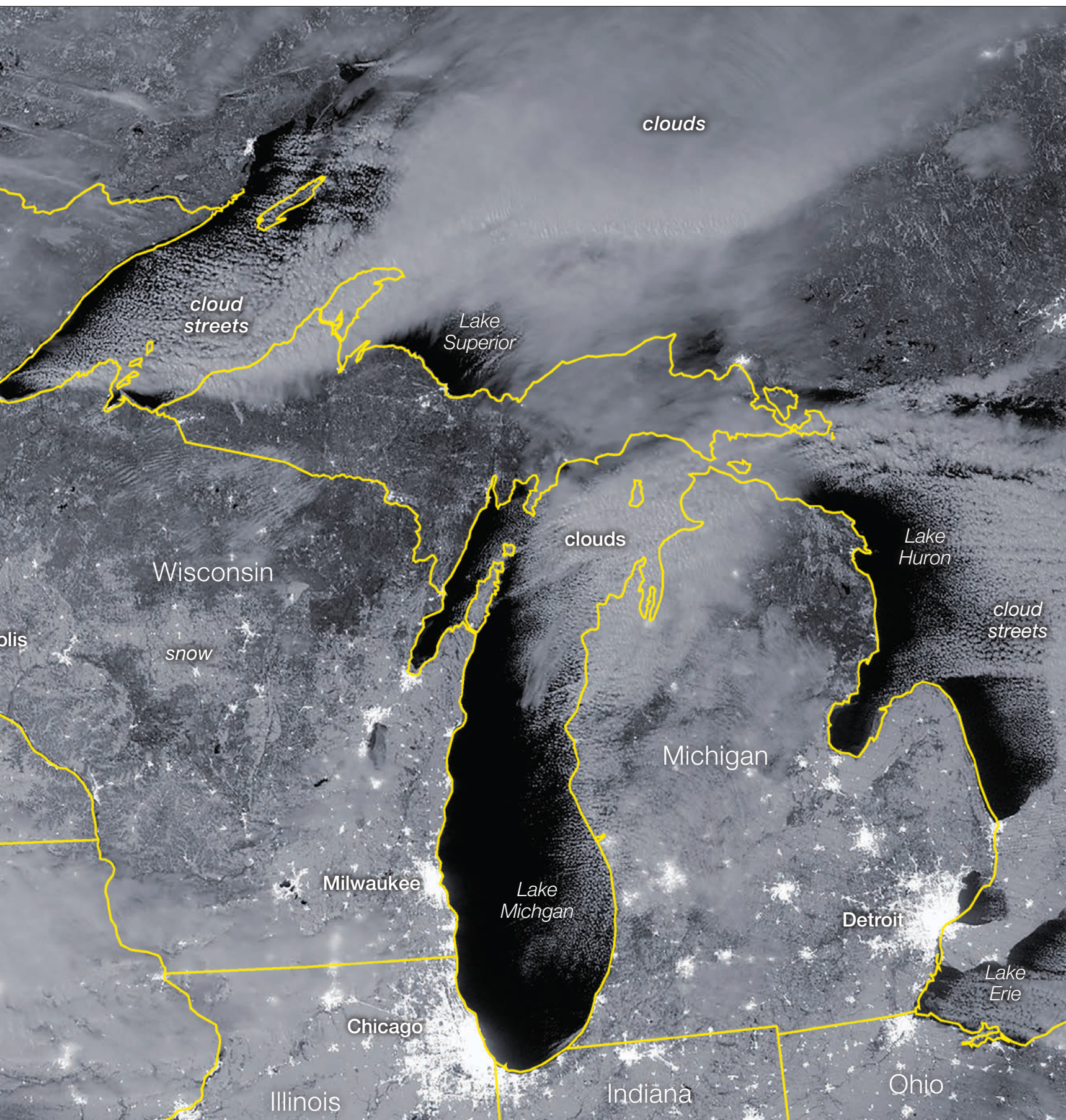


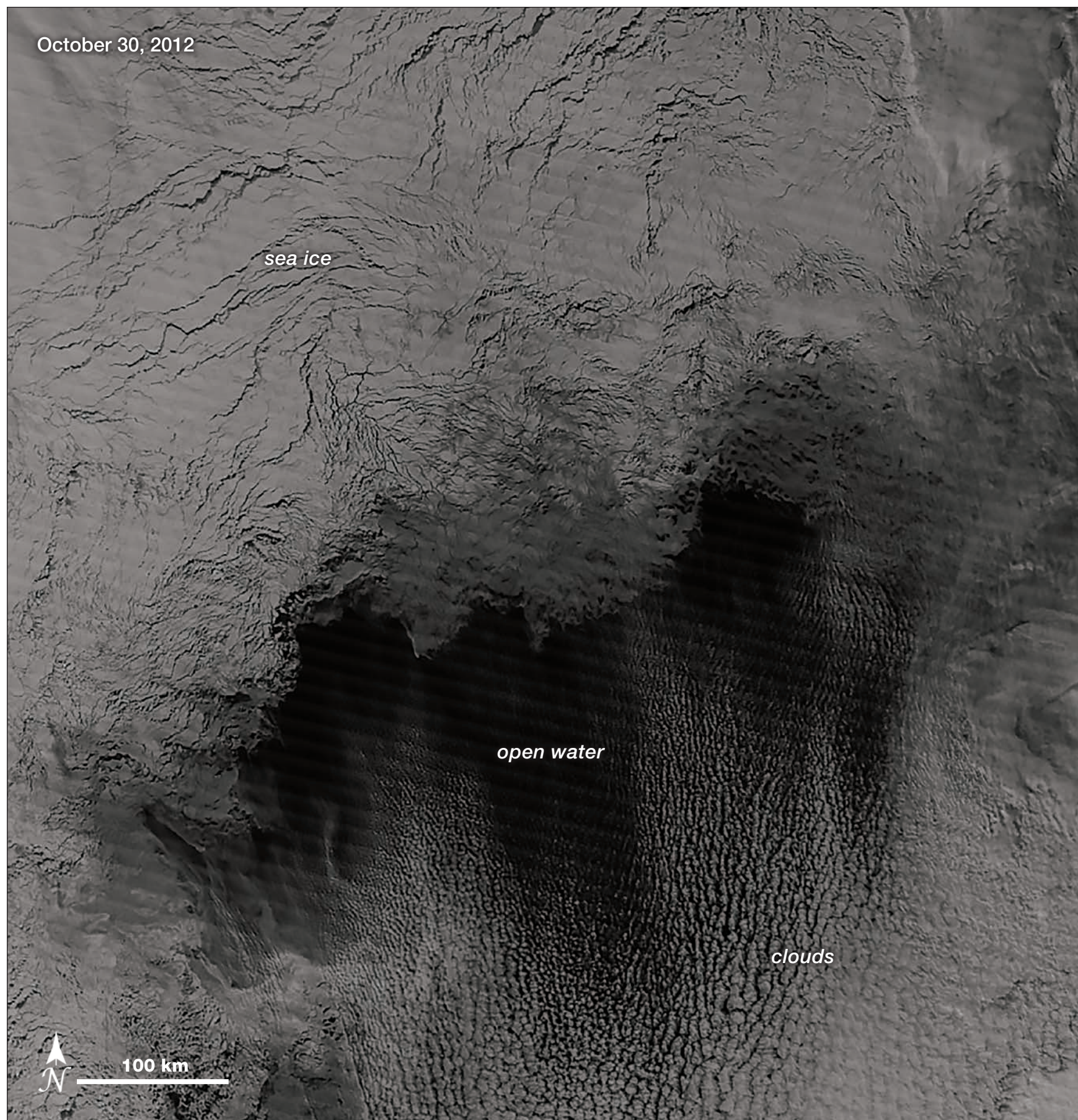
Snow and Ice

Snow Cover—Great Lakes Region, United States

An Arctic air mass brought snow to communities around the Great Lakes on December 14, 2016. The lake-effect snow came on the heels of an earlier accumulation that dumped several feet of snow in some areas the week before. The VIIRS DNB on the Suomi NPP satellite captured this image on December 14. The instrument can detect faint light sources—in this case, the reflection of the full Moon on the fresh snow. Clouds blur the landscape in the bottom left part of the image. Parallel rows of clouds, called cloud streets, appear above Lake Huron and Lake Superior. These clouds are created by cold, dry air blowing over a lake and accumulating water vapor.



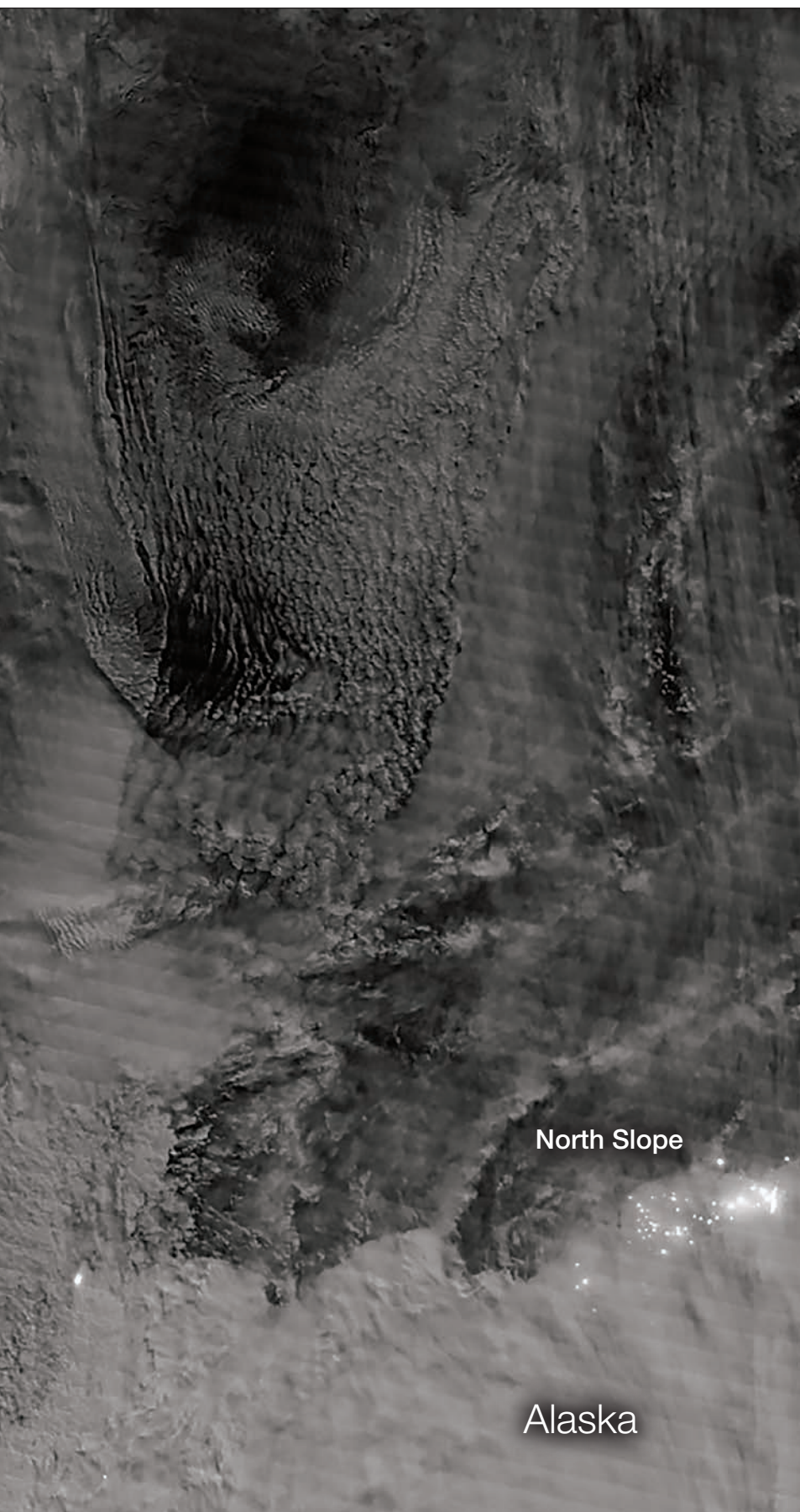






Polar Darkness—The Arctic

Scientists watched the Arctic with particular interest in the summer of 2012 when the areal extent of Arctic sea ice set a new record low. The behavior of sea ice following such a low extent also interests scientists, but as Arctic sea ice was advancing in the autumn of 2012, so was polar darkness. Fortunately, the VIIRS DNB on the Suomi NPP satellite can see in the dark and acquired this nighttime view of sea ice north of Russia and Alaska on October 30, 2012. During polar darkness (i.e., winter months), VIIRS DNB data allow scientists to observe sea ice formation and snow cover extent at the highest latitudes and to identify clear water where ships can safely navigate.

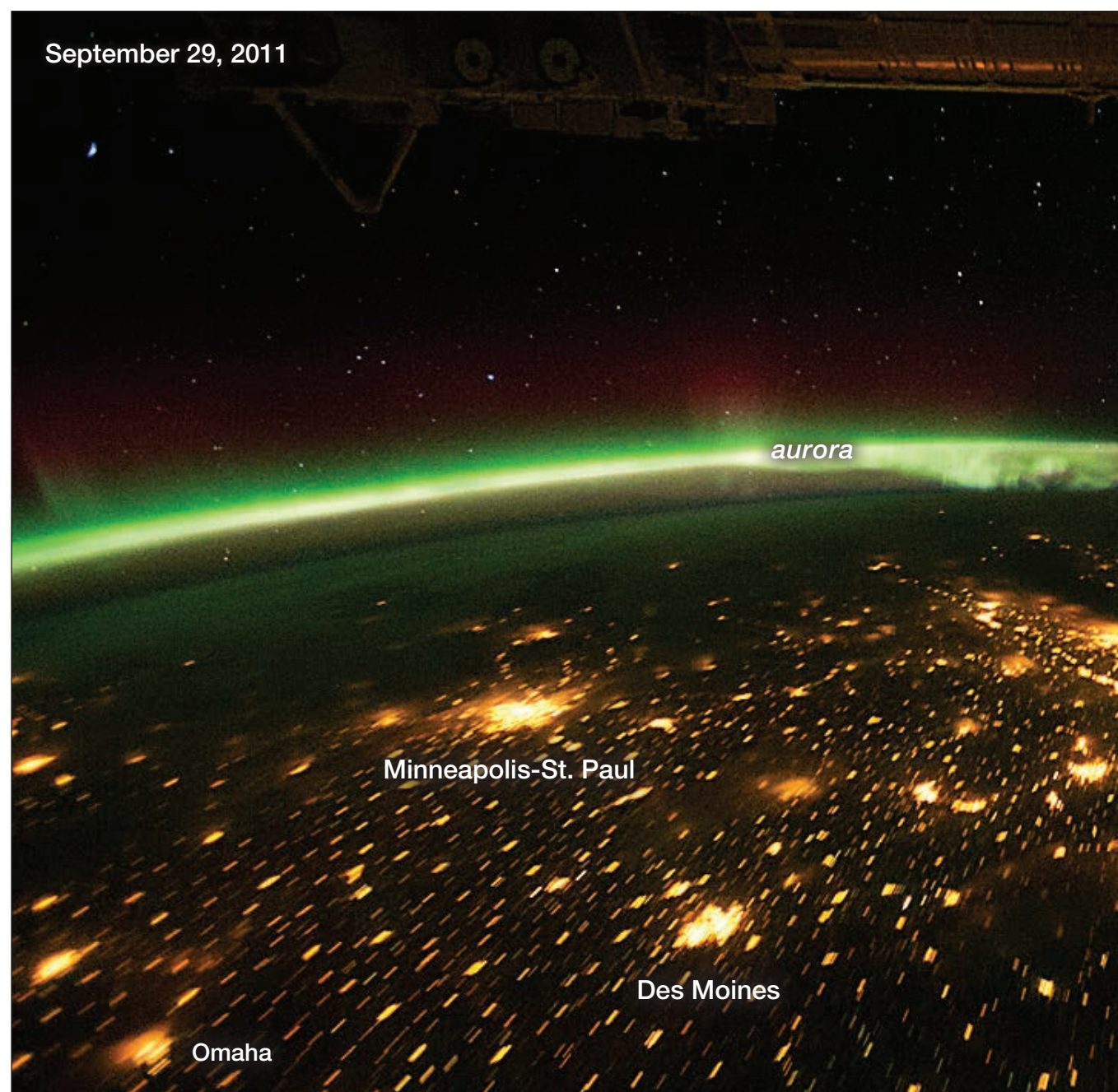




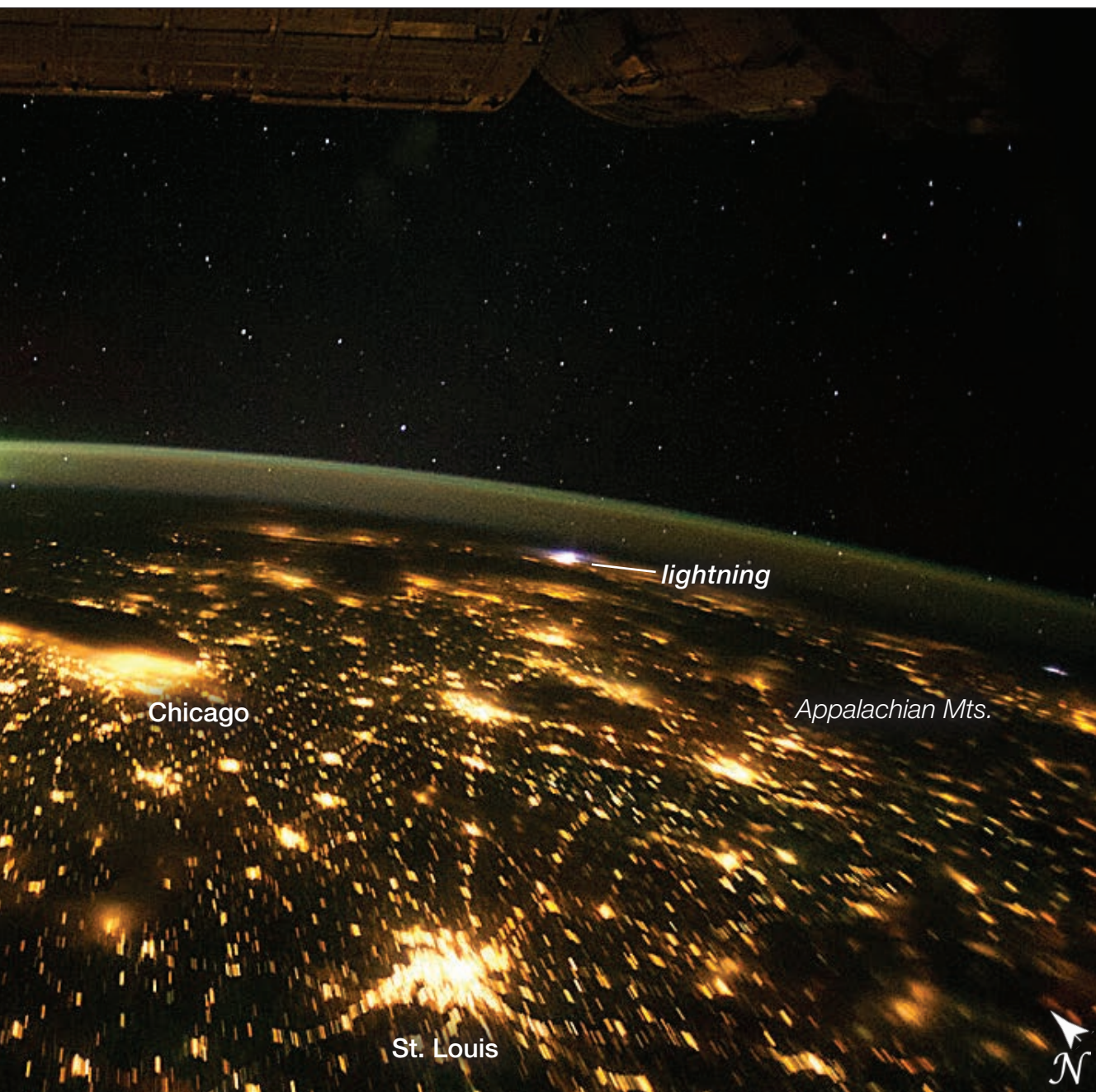
Auroras

Aurora Borealis—Midwestern United States

When viewed from the ISS, the night skies are illuminated with light from many sources. For example, the Midwestern United States presents a nighttime appearance when viewed from orbit, not unlike a patchwork quilt. The artificial light from human settlements appears with a characteristic yellow tinge, and the green light of the



aurora borealis shines brightly—even seeming to reflect off Earth’s surface in Canada—in this view from the ISS taken on September 29, 2011. A small white patch of light is almost certainly lightning from a storm on the East Coast (image right). Part of the ISS appears across the top of the image.





Looking Down at the Aurora—Northern Hemisphere

Even Earth's skies were celebrating St. Patrick's Day on March 17, 2015, as a severe geomagnetic storm—the strongest of the past decade—painted the sky with green, red, and blue auroras from New Zealand to Alaska. The storm conditions provided a fantastic opportunity for aurora viewing from above. On Sunday, March 15, a coronal mass ejection released large amounts of plasma and an accompanying magnetic field from the Sun, causing the aurora borealis to reach as far south as the central and southern United States, as shown on the image at right. The VIIRS DNB on the Suomi NPP satellite acquired this view of the aurora around 1:30 a.m. Eastern Daylight Time on March 18, 2015. Auroras appear as white streaks over Hudson Bay, southern Canada, and the northern United States.